

Thibault Pautrel

Postdoctoral researcher in applied mathematics and computer science
Riemannian optimization and federated learning

thibault.pautrel@centralesupelec.fr • +33 6 37 34 00 55 • thibault-pautrel.github.io



Born 17 August 1994 in Le Mans, France — French citizen

PROFESSIONAL EXPERIENCE

- | | |
|------------------------|--|
| 2025–present | Postdoctoral researcher
CentraleSupélec, Laboratory of Signals and Systems (L2S) — Gif-sur-Yvette
<i>Project:</i> Federated learning on covariance matrices for pre-trained models on health data.
<i>Supervision:</i> Florent Bouchard. |
| 2025
6 months | Research internship (MSc)
LISTIC, Université Savoie Mont Blanc — Annecy
<i>Project:</i> Riemannian Federated Learning on remote sensing data.
<i>Supervision:</i> Guillaume Ginolhac, Ammar Mian, Florent Bouchard. |
| 2021–2024
Full time | Mathematics teacher (<i>agrégé</i>)
French national education system
Teaching in secondary schools. Weekly oral examinations in undergraduate preparatory classes (BCPST, PCSI, MP). |
| 2018–2021 | PhD in Mathematics
Université de Rennes 1, IRMAR
<i>Title:</i> Some contributions to the study of the universality of the zeros of random trigonometric functions (defended 10 January 2022).
<i>Advisors:</i> Jürgen Angst and Guillaume Poly.
<i>Funding:</i> Doctoral contract with a teaching assignment (64 h/year). |
| 2018
5 months | Research internship (MSc)
LAMA, Université Savoie Mont Blanc — Chambéry
<i>Project:</i> Mean-reflected backward stochastic differential equations.
<i>Supervision:</i> Philippe Briand. |

PUBLICATIONS

Computer science (artificial intelligence & geometric learning)

- | | |
|-------------------|---|
| Keywords | Federated learning, Riemannian manifolds, SPD matrices, geometric deep learning, optimization on manifolds, signal processing. |
| [C3]
Submitted | T. Pautrel , F. Bouchard, A. Mian, G. Ginolhac. <i>Personalised federated learning for Riemannian and Euclidean EEG decoding</i> . Submitted to ICASSP 2027. arXiv:2609.29037 . |
| [C2]
Accepted | T. Pautrel , F. Portier. <i>Riemannian stochastic optimization for sufficient dimension reduction</i> . ICML 2026 |
| [C1]
Accepted | T. Pautrel , F. Bouchard, A. Mian, G. Ginolhac. <i>FedSPDnet: geometry-aware federated deep learning with SPDnet</i> . EuSIPCO 2026. |

Mathematics (stochastic processes & asymptotic analysis)

- | | |
|-------------------|--|
| Keywords | Nodal sets, random trigonometric polynomials, stationary Gaussian processes, strong dependence, universality, Kac–Rice formulas. |
| [R2]
To appear | J. Angst, T. Pautrel , G. Poly. <i>Global universality of the expected number of zeros of non-analytic</i> |

random signals. INdAM–Springer Proceedings, 2025.

[R1]
Published

J. Angst, **T. Pautrel**, G. Poly. *Real zeros of random trigonometric polynomials with dependent coefficients*. Transactions of the American Mathematical Society **375** (2022), 7209–7260.

[R0]
Published

T. Pautrel. *New asymptotics for the mean number of zeros of random trigonometric polynomials with strongly dependent Gaussian coefficients*. Electronic Communications in Probability **25** (2020), no. 36.

TEACHING

University teaching

2026–2027
6 h TD-eq.

Labs: Machine Learning for Data Science

ENSAI — MSc Smart Data

17 students.

2025–2027
33 h TD-eq.

Labs: Introduction to C++ programming

IUT Orsay — 1st year (R101-B)

15 students, 5.5 h per week.

2025–2026
12 h TD-eq.

Tutorials and labs: Machine Learning

CentraleSupélec — 2nd year

102 students, taught in English, 8 sessions of 1.5 h.

2025–2026

Lectures and tutorials: Statistics and Learning

CentraleSupélec — 1st year

≈30 students, taught in French and English (S2).

Lectures: $2 \times 1.5 \text{ h} = 4.5 \text{ h TD-eq.} + 15 \text{ h of tutorials.}$

2025–2026
1.5 h TD-eq.

Tutorials: Digital image processing

CentraleSupélec — 2nd year

Taught in French.

2018–2021
48 h TD-eq.

Tutorials: Analysis and probability

Université de Rennes 1 — 2nd year, Mathematics

≈30 students, 4 h per week.

2018–2021
16 h TD-eq.

Tutorials: Probability

Université de Rennes 1 — 1st year, Biology

≈30 students, 2 h per week over 8 weeks.

Supervision and examination boards

2026

Master's thesis examination board

ENSAI — MSc Smart Data

2023–2025

Examiner, CentraleSupélec entrance examination

Marker for the written MP Mathematics 1 papers (2023, 2024, 2025).

Examiner for the Mathematics–Computer Science oral examinations (2023, 2024).

2019–2020

Supervision of undergraduate research projects

ENS Rennes — 1st year

Topic: Stein's method for normal approximation.

2018–2021

Mock oral examination board

Université de Rennes 1

Preparation for the CAPES and the *agrégation* in mathematics.

Oral examinations in undergraduate preparatory classes

2021–2024

Weekly oral examinations in mathematics

BCPST, PCSI, MP — various institutions

2016–2018 **Weekly oral examinations in mathematics**
MP, MP* — Lycée Chateaubriand, Rennes

Secondary school teaching

2021–2024 **Mathematics teacher (*agrégé*)**
Full time Secondary schools

EDUCATION

2024–2025 **MSc Smart Data**
ENSAI, Rennes — with highest honours
Specialisation: machine learning, deep learning, dimension reduction, functional data analysis.

2017–2018 **MSc in Probability and Statistics (“Aléatoire”)**
Université de Rennes 1 — with honours
Specialisation: stochastic processes and stochastic calculus, statistics of processes, dynamical systems, parametric and non-parametric statistics, Malliavin calculus and Stein’s method

2016–2017 **Preparation for the *agrégation* in mathematics**
Université de Rennes 1 — with honours

2015–2016 **MSc (first year) in pure mathematics**
Université de Rennes 1 — with honours

2014–2015 **BSc in mathematics**
Université de Rennes 1 — with honours

2012–2014 **Undergraduate preparatory classes, MPSI – MP***
Lycée Chateaubriand, Rennes

COMPETITIVE EXAMINATIONS AND DISTINCTIONS

2026 **National qualification for Associate Professor positions**
CNU section 26 — applied mathematics and applications of mathematics

2017 ***Agrégation externe* in mathematics**
Probability–statistics option — rank: 118

SERVICE

2026 **Reviewer**
ICML 2026 — International Conference on Machine Learning

2019–2021 **Co-organiser of the doctoral seminar**
“Gaussbusters” seminar — IRMAR, Rennes
Weekly seminar of the PhD students in probability and statistics.

TECHNICAL SKILLS

Languages Python (NumPy, PyTorch, scikit-learn), R, SQL, C++, OCaml, Scilab
Computing Distributed computing (Spark), parallel computing, HTCondor scheduler
Tools Git, $\text{\LaTeX}$, Unix/Bash

LANGUAGES

French Native
English Fluent (C2) *Certificate of Proficiency in English (Cambridge, 2013)*
Spanish Intermediate (B2) *Read, written, spoken*
Breton Beginner